

Sumit Chhajed

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EXPERIENCE

Tata Advanced Systems Limited

Mumbai

Software Engineering Intern

Jun 2026 – Present

- Developing SPIDER, a full-stack security and surveillance platform for defense infrastructure, focusing on end-to-end application functionalities, real-time asset monitoring, and high-throughput data processing.
- Designing scalable backend architectures and integrating third-party APIs in a Linux environment to ensure low-latency system performance and optimized concurrent data pipelines.

Webgile Solutions

Nagpur, India

Data Analyst Intern (Python & SQL)

Dec 2025 – Jan 2026

- Engineered and optimized ETL data transformation pipelines using Python (Pandas, NumPy) and SQL, executing data mapping and processing large datasets for statistical analysis and system reliability.
- Constructed complex MSSQL and PostgreSQL queries to track KPIs, utilizing data-driven methods, performance tuning, and statistical concepts to support actionable quantitative decision-making.

PROJECTS

High Performance Remote Desktop Architecture

- Architected a cross-platform, production-grade remote desktop clone using Rust, Flutter, and WebSockets for low-latency, high-throughput concurrent data transmission.
- Designed a scalable signaling server matching top-tier repositories for seamless real-time connectivity across the full development lifecycle.

DeepHider Steganography & Cryptography Engine

- Engineered a custom 32-bit bit-plane slicing PC desktop application in C/C++ to losslessly embed high-resolution image payloads with optimal efficiency.
- Implemented a secure XOR stream cipher with SHA-256 key expansion, optimizing memory usage and execution speed using OOP principles.

IoT Based Smart Helmet for Delivery Workforce Safety

- Developed an IoT smart helmet (Raspberry Pi 4, ESP32) featuring alcohol monitoring, drowsiness detection, and fall detection architectures.
- Integrated Firebase, GPS tracking, and Twilio APIs for real-time safety monitoring, low-latency emergency alerts, and concurrent vehicle access control.

EDUCATION

Thadomal Shahani Engineering College

Mumbai

B.E. in Electronics and Telecommunication; CGPA: 8.20 / SGPA: 9.14

Aug 2023 – 2027

- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming (C++), Database Management Systems

Nahata College — XII Grade HSC

2021 – 2023

Tapti Public School — X Grade CBSE

2021

SKILLS & INTERESTS

- **Languages:** C/C++, Python, SQL (MSSQL, PostgreSQL), Rust, VHDL, JavaScript, HTML/CSS
- **Libraries & Frameworks:** Scikit-learn, Pandas, NumPy, Flutter, WebSockets, OpenCV, MediaPipe, Firebase
- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, AWS, Xilinx ISE, TCP/UDP Protocols, DevOps
- **Core Competencies:** Full Stack Development, ETL Development, Data Mapping, Object-Oriented Programming (OOP), Data Structures & Algorithms, Scalable Solutions, Ownership Attitude
- **Achievements & Leadership:** Technical Head @ ISTE TSEC (2025–2026); 2nd Runner Up at Tech-athon (90+ teams) & TSEC Hardware Exhibition; Solved 100+ LeetCode DSA problems; 96.14 percentile MHT CET.